Topic 5 – Factors influencing product development

5.1 User centred design factors

Spec: The importance and influence of user centred design in ensuring products are fit-for-purpose and meet the criteria of specifications when designing, making and evaluating in relation to:; User needs, wants and values; purpose; functionality; innovation; authenticity

Key facts

• **Needs vs wants?** Needs are essential requirements; wants are desirable extras.
• **Give three examples of user values.** Sustainability/ethics, cultural or religious beliefs, value for money, brand image, safety, social status.
• **How can designers find user needs?** Interviews, questionnaires, observation, focus groups, user trials/testing, personas, empathy tools, anthropometric data, analysing existing products.
• **Why is UCD iterative?** Ideas are prototyped, tested with users and refined repeatedly, so problems are found early and the final product better meets needs.
• **Two UCD considerations other than needs, wants and values?** Purpose (intended use/environment) and functionality (how the user interacts, ergonomics, safety). Also innovation and authenticity.
• **Limitations of UCD?** Users may not know what they want or can't imagine new technology; time-consuming and costly; small samples may not represent all users; designing for everyone can over-complicate products.

Factor	What the designer considers	Example (multi-purpose baby chair, 2021 Q7)
User needs	Essential requirements for use, safety, size, abilities, disabilities	Safety harness, lockable wheels, stable base
User wants	Desirable features, aesthetics, extras	Activity centre, colour choice, reversible padding
User values	Beliefs, ethics, culture, price, sustainability, brand	Value for money (lasts birth–3 years), non-toxic materials
Purpose	Intended use, user, environment, lifespan	Feeding and play at home; must be cleaned often
Functionality	How well it works, ease of use, ergonomics, safety, maintenance	8 height positions, folding frame, removable tray
Innovation	New materials, technology or approach that improves performance, cost or usability	Padded reducer converting a newborn recliner into a toddler seat
Authenticity	A genuine, original solution, true to the users and the brand; every stage of the design process valued	Designed through real family testing, not a copy of a rival

Key terms

User centred design (UCD)	An iterative design approach where the needs, wants and values of the end users are the focus at every stage – research, specification, design, prototyping and evaluation.
Needs	What the user must have for the product to work for them (essential requirements, e.g. a child seat must be safe).
Wants	Desirable features the user would like but does not strictly need (e.g. a particular colour, extra features).
Values	The user's beliefs and priorities that affect choice – ethical, environmental, cultural, religious, brand loyalty, price.
Purpose	What the product is for – its intended use, user and environment.
Functionality	How well the product performs its functions and how the user interacts with it (usability, ergonomics, safety).
Innovation	Introducing something new or improved – new materials, technologies, methods or approaches to solve a problem better.
Authenticity	The design is genuine, original and true to the user's real needs and the brand – an honest approach where the user is the true focus, not a copy or gimmick.
Fit for purpose	The product does the job it was designed for, in its intended environment, for its intended users, as defined by the specification.

Exam tip (2024 Q6b): The examiners wanted two **other** considerations (purpose, functionality, innovation, authenticity) each explained with **two linked points**. Repeating needs, wants and values from the stem scored nothing.

5.2 Anthropometrics and ergonomics

Spec: Sources and applications of anthropometric data; Ergonomic factors for a designer to consider when developing products and environments with which humans interact

Key facts

• **Anthropometrics vs ergonomics?** Anthropometrics = body measurements (data). Ergonomics = applying that data and other human factors so products are comfortable, efficient and safe.
• **Why design for the 5th to 95th percentile?** It fits 90% of users without making products impractical or too costly for the extremes.
• **Which percentile for a door height? For a shelf reach?** Door height: 95th+ percentile stature (tall people). Shelf/reach: 5th percentile reach (shorter people can still reach).
• **Two official sources of anthropometric data?** British Standards Institution (BS EN ISO 7250), ISO, Department for Business (formerly DTI) 'Adultdata/Childata', European sizing surveys, WHO, PeopleSize software, Pheasant's 'Bodyspace'.
• **Ergonomic factors to consider?** Size and fit, reach, posture, comfort, grip, weight/force needed, controls layout and feedback, visibility, lighting, noise, temperature, safety, cognitive load, accessibility.
• **Uses of anthropometric data in clothing?** Creating sizing systems, positioning pockets/zips, predicting stock ratios for each size from the normal distribution, maximising fit and freedom of movement.
• **What is inclusive design?** Designing products usable by as many people as possible, including older and disabled users, without the need for adaptation.

Design situation	Use	Example
Clearance (people must fit through/under)	95th percentile (large)	Door heights, car headroom, seat widths
Reach (people must reach)	5th percentile (small)	High shelves, controls on a dashboard, pedal distance
Adjustable products	Range 5th–95th	Office chairs, car seats, spin bikes (2024 Q11), clothing sizes

Key terms

Anthropometrics	The study and measurement of the human body – sizes, reach, strength and proportions – used to design products that fit people.
Ergonomics	The study of how people interact with products, equipment and environments, so they are comfortable, efficient and safe to use.
Percentile	The percentage of a population below a given measurement. The 95th percentile stature means 95% of people are shorter.
5th–95th percentile	The range designers usually design for, accommodating 90% of the target population and excluding the extremes.
Static data	Measurements of the body at rest – e.g. height, hand length, sitting height.
Dynamic data	Measurements during movement – e.g. reach, grip strength, range of joint movement.
Normal distribution	Bell-shaped curve showing that most people are near the average and fewer at the extremes.
Outlier	A person whose measurements fall outside the range designed for (below 5th or above 95th percentile).

Exam tip (2025 Q6b): Uses of anthropometrics in clothing went beyond "it fits": sizing systems, **predicting demand and stock ratios** for each size from the normal distribution, and positioning pockets, zips and fastenings.

5.3 Aesthetics and form versus function

Spec: The influence of aesthetics, ergonomics and anthropometrics on the design, development and manufacture of products; Form over function; Form follows function

Key facts

• **Who coined 'form follows function' and which movement adopted it?** Architect Louis Sullivan (1896); adopted by the Bauhaus and Modernism.
• **Give an example of form over function.** Philippe Starck's Juicy Salif lemon squeezer – sculptural but juice runs down your hand; Loewy's streamlined pencil sharpener.
• **Give an example of form follows function.** Marianne Brandt's Kandem lamp, a Swiss army knife, a hammer, a Brompton folding bike, a games controller shaped by the hands.
• **Aesthetic elements to discuss?** Shape/form, colour, texture, pattern, proportion, symmetry, line, contrast, harmony, materials and finish, style/fashion, sound, weight.
• **Where might aesthetics and ergonomics conflict in a car interior?** Touchscreens look sleek but lack tactile feedback; low bucket seats look sporty but are hard to get out of; gloss trim looks premium but causes glare and fingerprints.

Key terms

Aesthetics	How a product appeals to the senses – look (form, colour, proportion, pattern), feel (texture, weight), sound, smell and taste.
Form	The shape, appearance and style of a product.
Function	What the product does and how well it does it.
Form follows function	The shape of a product should come from its purpose and use (Louis Sullivan; Bauhaus). Decoration is minimised.
Form over function	Appearance is prioritised over practicality – the product is designed to look good or make a statement, sometimes at the expense of how well it works (e.g. Juicy Salif, Loewy's pencil sharpener).
Proportion	The relationship of sizes of parts to each other and the whole – e.g. golden ratio 1:1.618.

Exam tip (2025 Q9): Discuss genuine **conflicts** between aesthetics and ergonomics in depth – don't just list car features.

5.4 Design movements and designers

Spec: Arts and Crafts – William Morris; Art Nouveau – Charles Rennie Mackintosh; Bauhaus Modernist – Marianne Brandt; Art Deco – Eileen Gray; Post Modernism – Philippe Starck; Streamlining – Raymond Loewy; Memphis – Ettore Sottsass

Key facts

• **William Morris quote?** 'Have nothing in your houses that you do not know to be useful, or believe to be beautiful.'
• **Key Morris works?** Red House (1859, with Philip Webb), Strawberry Thief textile (1883), Trellis and Daisy wallpapers, Morris & Co (1861), Kelmscott Press, the Morris adjustable chair.
• **Contradiction in Arts and Crafts?** It aimed for good design for ordinary people, but hand-crafted goods were expensive so only the wealthy could afford them.
• **Key Mackintosh works?** Glasgow School of Art (1897–1909), Hill House (1902–04), Willow Tea Rooms, high-backed Argyle and Hill House ladder-back chairs, the stylised Glasgow rose.
• **Features of Mackintosh's Glasgow Style?** Elongated verticals, high-backed chairs, grids/squares and geometry combined with stylised roses, Japanese influence, black/white with purple/pink accents, holistic design of building and contents.
• **Marianne Brandt – key products?** MT49 tea infuser/teapot (1924; brass, silver, ebony – circles and semicircles) and the Kandem bedside lamp (1928) – mass-produced, adjustable, form follows function.
• **Bauhaus dates, founder and principles?** 1919–1933, Walter Gropius; form follows function, geometric shapes, primary colours, minimal ornament, new materials (tubular steel, glass), designing for mass production, unity of art and craft.
• **Eileen Gray – key products?** E-1027 adjustable side table (1927, tubular steel + glass, height-adjustable), Bibendum chair (1926), Dragon chair (lacquer), villa E-1027; lacquer screens.
• **Art Deco features?** Geometric zig-zags, chevrons, sunbursts/fans, stepped ziggurat forms, symmetry, verticality, luxurious materials (marble, chrome, lacquer, exotic woods), bold colour, Egyptian/Aztec influences, 'ocean liner' and 'Hollywood' glamour.
• **Philippe Starck – key products?** Juicy Salif lemon squeezer (1990, Alessi, cast aluminium), Louis Ghost chair (2002, Kartell, polycarbonate), Hot Bertaa kettle, Dr Glob chair; 'democratic design' – good design for many.
• **Raymond Loewy – key products?** Coldspot refrigerator (1934), pencil sharpener (1933), Pennsylvania Railroad S1/GG1 locomotive styling, Greyhound Scenicruiser, Studebaker cars, Lucky Strike pack, Shell and Exxon logos.
• **Loewy's famous ideas?** 'Ugliness does not sell'; MAYA – Most Advanced Yet Acceptable; known as the father of industrial design.
• **Ettore Sottsass – key products?** Carlton bookcase/room divider (1981), Casablanca sideboard, Tahiti lamp; earlier Olivetti Valentine typewriter (1969).
• **Memphis features?** Bright clashing colours, plastic laminate (Abet Laminati) and terrazzo, squiggles and geometric patterns, asymmetric playful forms, kitsch and Pop Art references, deliberately challenging 'good taste'.

Movement	When	Lines	Ornament	Materials	Production
Arts & Crafts	1860–1910	Simple, natural pattern	Nature-based, restrained	Natural, local	Hand-crafted
Art Nouveau	1890–1910	Flowing whiplash curves (Mackintosh: geometric verticals)	Organic	Iron, glass, wood	Craft + some industry
Art Deco	1920–1939	Geometric, symmetrical, stepped	Rich, geometric	Luxurious + chrome, Bakelite	Craft + industry
Bauhaus	1919–1933	Pure geometry	None	Steel, glass	Mass production
Streamlining	1930s–50s	Curved teardrop, horizontal lines	Speed lines only	Chrome, pressed steel, Bakelite	Mass production
Memphis	1981–87	Angular, asymmetric	Bold pattern	Laminates, plastics	Small batch
Post Modernism	1970s–	Playful, sculptural	Symbolic, witty	Polymers, cast metals	Mass production

Key terms

Arts and Crafts	c.1860–1910, Britain. Reaction against poor-quality industrial mass production; championed handcraft, honest construction, natural materials and nature-inspired pattern. Key figure: William Morris (with John Ruskin).
Art Nouveau	c.1890–1910. 'New art' using flowing organic 'whiplash' curves, plant and insect forms and new materials such as iron and glass. Mackintosh's Glasgow Style added geometry and Japanese influence.
Bauhaus	German design school 1919–1933 (Weimar, Dessau, Berlin), founded by Walter Gropius. Form follows function, geometric simplicity, unity of art and craft, design for industrial mass production.
Art Deco	1920s–1930s, from the 1925 Paris Exposition. Glamorous, luxurious, geometric – zig-zags, sunbursts, chevrons, symmetry, stepped forms, Egyptian/Aztec influences, expensive materials and new materials such as chrome and Bakelite.
Streamlining (Streamline Moderne)	1930s–1950s USA. Aerodynamic teardrop shapes, smooth rounded forms and horizontal speed lines applied to products to make them look modern and sell – often form over function. Led to Googie architecture.
Post Modernism	1970s onwards. Rejected Modernism's strict functionalism; embraced humour, colour, symbolism, irony and emotion. Philippe Starck.
Memphis	Milan design group 1981–1987 led by Ettore Sottsass. Bold clashing colours, plastic laminates, geometric pattern-on-pattern, kitsch – challenged 'good taste'.
Googie	1950s–60s Californian architecture from Streamline Moderne: upswept cantilevered roofs, boomerangs, starbursts, neon and space-age imagery to attract motorists.
MAYA	Loewy's principle 'Most Advanced Yet Acceptable' – push design forward but not so far that consumers reject it.